

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think>

First finds initial distance : $3 * 60 = 180$ miles .

Then in next 2 hrs he covers 0 miles .

Next half-hour at 30 mph covers $30 / 2 = 15$ miles .

Remaining time is $4 - 2 - 0.5 = 1.5$ hrs .

So last leg covers $1.5 * 80 = 120$ miles .

So total distance covered is $15 + 120 = 135$ miles .

So distance from home is $180 - 135 = 45$ miles .

<answer> 45 </answer>

